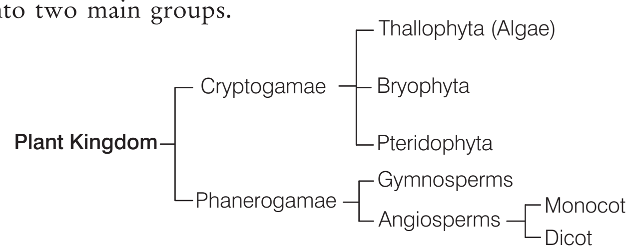
- **Nomenclature** means providing distinct and proper names to organisms, so that they can be easily recognised and differentiated from others. The two codes for nomenclature are
  - **ICBN** International Code of Botanical Nomenclature
  - **ICZN** International Code of Zoological Nomenclature
- **Binomial nomenclature** was given by Linnaeus. It is the system of nomenclature using two terms, the first indicating the **genus** and the second one indicating the **species**, e.g. *Mangifera indica* (mango).
  - The categories used in classification are kingdom, phylum, class, order, family, genus and species.
  - Genus and species terms were proposed by **John Ray**.
  - The smallest unit of classification is species.
- **Living organisms** have been classified by different scientists. Some of the main classification systems are:
  (*i*) Two system classification was given by **Linnaeus** in 1758. It was the first classification. He divided all the living organisms into two kingdoms. These are Plantae and Animalia.
  (*ii*) **Copeland** designed four kingdom classification system in 1956.
  (*iii*) **Whittaker** gave five kingdom classification system in 1969.

CLASSIFICATION OF PLANTS

Many systems for classifying the plant kingdom have been formulated by various researchers and scientists, from time to time. Scientist **Eichler** has classified the plant kingdom into two main groups.



1. Cryptogamae

These plants never bear flowers and seeds. These are the most primitive. This division includes three main groups:

(i) Thallophyta

These are the simplest and primitive plants. Plant body is an undifferentiated mass of cells called as **thallus**. They lack vascular tissues.

This group includes algae and lichens (a symbiotic combination of an alga with a fungus).

(a) **Algae** These are plant-like organisms that are usually photosynthetic and aquatic in nature.
These are chlorophyll containing thallophytes, in which vascular tissue (xylem and phloem) are absent.
Some examples are as follow:
- *Chlorella* used as component of sewage oxidation tanks.
- Antibiotic chlorellin is obtained from *Chlorella.*
- Iodine is obtained from *Laminaria.*
- Agar is obtained from *Gelidium* and *Gracilaria*.
- *Spirogyra* is also called as **pond silk**.

(b) **Fungi** These are non-vascular, heterotrophic, spore forming eukaryotic organisms.
They have stored food material as glycogen. Cell wall is made up of chitin or fungal cellulose.
Some examples are as follow:
- *Albugo candida* – White rust of crucifers
- *Puccinia graminis tritici* – Black rust of wheat
- *Rhizopus stolonifer* – Fumatic acid
- *Aspergillus* – Produces aflatoxin (natural mycotoxin)
- *Agaricus* – Mushroom, these have filamentous structure called **hyphae**.
- In Whittaker's five kingdom classification, fungi has been placed in a separate kingdom – Fungi. It is mainly because fungi do not have chlorophyll.

(c) **Lichens** Symbiotic relationship between algae and fungi. These are the indicators of water pollution.

(ii) Bryophyta

These are called as the amphibians of plant kingdom.

(a) They do not possess vascular tissue.
(b) Roots are generally absent in them. They are present in damp and shady places.
(c) Some examples are as follow:
- *Riccia, Marchantia*, *Funaria*.
- *Zoopsis* is the smallest bryophyte.
- *Dawsonia* is the largest moss.
- *Sphagnum* is employed for dressing of wound.

(iii) Pteridophyta

The term 'Pteridophyta' comes from Greek word *Pteris* means fern and *phyton* means plant.

(a) These are seedless vascular plants that have plant body differentiated into true roots, stems and leaves. Secondary growth is absent in them.
(b) Some examples are as follow:
- *Selaginella, Equisetum.*
- *Dryopteris* (male shield fern).
- *Azolla, Marsilea, Salvinia* are aquatic ferns.
- *Adiantum* (maiden hair fern).
- *Cyathium* (tree fern).

2. Phanerogamae

These are also known as **spermatophytes**. They include those plants, which produce seeds. These are divided into two major types:

(i) Gymnosperms

The term 'Gymnosperm' comes from Greek word *Gymno* means naked and *sperma* means seed. These are vascular plants. Vascular plants means xylem and phloem are present in them for transportation of water and food, respectively.

Gymnosperms do not have outer covering around their seeds, i.e. their seeds are naked. Pollination type is anemophily (pollination by wind).

Some examples are as follow:

- *Cycas* (sagopalm)-*Nostoc* /*Anabaena* show symbiotic association with *Cycas* (most primitive gymnosperm).
- *Pinus* – Resin and terpentine obtained from *Pinus chilgoza* is obtained from *Pinus*.
- *Ginkgo* (maiden hair tree) – Living fossil.
- *Ephedra* – Drug ephedrine is obtained from the stem of *Ephedra*.
- *Sequoia gigantia* (red-wood tree), tallest (largest tree in term of total volume).
- *Rhynia* (most primitive gymnosperms).

(ii) Angiosperms

These are seed bearing, flowering vascular plants. Their seeds are enclosed within fruits. Double fertilisation takes place in angiosperms. Flowers are generally bisexual, e.g. mustard plant (*Brassica campestris*).

They are further divided into two types:

(a) **Monocots** These are angiospermic or flowering plants, which are characterised by the presence of single cotyledon in their seeds, e.g. maize, grass, etc.

(b) **Dicots** These are angiospermic flowering plants characterised by the presence of two cotyledons in their seeds, e.g. pea, rose, cotton, sunflower, etc.

CLASSIFICATION OF ANIMALS

Animals are classified in a variety of ways. This helps scientists to study the relationships in animal groups and to see the whole animal family as it has developed through time. Phylum is the grouping of different classes of animals, which have some common characteristic features like mode of nutrition, reproduction, locomotion, etc.

Storer and **Usinger** classified animals into following phylums:

Phylum–Protozoa

- These are unicellular eukaryotic organisms.
- In these, all the metabolic activities like digestion, respiration, excretion and reproduction take place in unicellular body.
- Respiration and excretion take place through diffusion, e.g. *Amoeba*, *Paramecium*, *Plasmodium* (malarial parasite).
- *Euglena* have characters of both animals and plants.
- Phylum–Protozoa has now been placed in kingdom–Protista.

Phylum–Porifera

- These are multicellular animals.
- These all are found in marine water and have porous body. The pores are called **ostia**.
- Their skeleton is made up of minute calcareous or silicon spicules, e.g. *Sycon*, sponge, etc.

Phylum–Coelenterata

- These are aquatic animals that possess thread-like structures called **tentacles** around the mouth, which help in holding the food.
- They have specialised cnidoblast cells to help in catching the food.
- *Hydra* has a tendency of regeneration of body organs, ***e.g.*** *Hydra*, jellyfish, sea anemone, etc.

Phylum– Platyhelminthes

- These animals are also called **flatworms**.
- Excretion takes place by flame cells.
- There is no skeleton, respiratory organ, circulatory system, etc., but alimentary canal is present.
- These are hermaphrodite animals. e.g. *Planaria*, liver fluke, tapeworm, etc.
- These organisms are found as internal parasites. For example, tapeworm is a common parasite found in intestine of human beings.

Phylum–Aschelminthes

- These are long cylindrical, unsegmented and unisexual worms, also called **Nematoda** or round worms.
- Their alimentary canal is complete, in which mouth and anus both are present.
- There is no circulatory and respiratory systems, but nervous system is well-developed.
- Excretion takes place through protonephridia.
- Most forms are parasitic, but some are free -living in soil and water, e.g. *Ascaris*, threadworm, etc.
- **Threadworm** is found mainly in the anus of child. Children feel itching and often vomit. Some children urinate on the bed at night.

- *Wuchereria bancrofti* (filaria) is human parasitic roundworm, which causes elephantiasis (filariasis) that spread through larvae. The **mosquito** (*Culex*) is the intermediate host.

Phylum–Annelida

- These are segmented worms.
- Their body is long, thin and soft.
- Alimentary canal is well-developed, nervous system is normal and the colour of the blood is red.
- They respire through skin, excretion occurs by nephridia.
- They move through setae made up of chitin, e.g. earthworm, *Neries*, leech, etc.
- Earthworm has four pair of hearts and it possess both male and female sex organ, i.e. it is hermaphrodite.
- Earthworm's body is sensitive for acidic, basic temperature and humidity.

Phylum–Arthropoda

- This is the largest phylum of Animalia.
- Jointed leg is their main feature. Their body is divided into three parts, i.e. head, thorax and abdomen. It is covered by chitinous exoskeleton.
- Circulatory system is open type.
- Cockroach’s heart has **13 chambers** and has spiracles.
- Trachea, book lungs, body surface are respiratory organs.
- Insects generally have six feet and four wings, e.g. cockroach, prawn, crab, bug, housefly, mosquito, bees, etc.
- Arachnids possess 8 legs, e.g. spider.
- Ant is a social animal, which reflects labour division.
- Termite is also a social animal, which lives in colony.
- Insect bite releases formic acid, which can cause itching, swelling, redness, etc.

Phylum–Mollusca

- Their body is soft unsegmented and divided into head, muscular foot and visceral hump.
- Body is covered by a hard calcareous shell.
- Their alimentary canal is well-developed.
- Respiration takes place through gills or ctenidia.
- Blood is blue due to the presence of haemocyanin.
- Excretion takes place through kidneys. e.g. *Pila, Aplysia* (sea rabbit), *Doris* (sea lemon), *Octopus* (devilfish), *Sepia* (cuttlefish).

Phylum–Echinodermata

- All the animals in this group are marine.
- They have water vascular system.
- Brain is not developed in nervous system.
- They have a special capacity of regeneration, e.g. starfish, sea urchin, sea cucumber, etc.

Phylum –Hemichordata

Hemichordata was earlier placed as a sub-phylum under the phylum-Chordata. But now, it is considered as a separate phylum (under non-chordata). These are also called

Phylum–Chordata

- They have notochord, a dorsal hollow tubular nerve cord and paired pharyngeal gill slits.
- This phylum is subdivided into two **subphylum**, i.e. Protochordata and Vertebrata.

Some main groups of phylum– Chordata are:

Group–Pisces

- These are aquatic animals (cold-blooded animals).
- Their heart pumps only impure blood and have two chambers.
- Class–Cyclostomata includes jawless fishes, e.g. lampreys.
- Cartilaginous fishes come under class–Chondrichthyes, e.g. Indian shark.
- Class–Osteichthyes includes bony fishes, e.g. sea-horse (*Hippocampus*)
- Respiration takes place through gills, gill slits, etc. e.g. *Scoliodon* (dogfish), *Torpedo,* etc.

Group–Tetrapoda

Group Tetrapoda classified into four clasess

(i) Amphibia

- These can live both on land and water. All these animals are cold-blooded or ectothermic.
- Ectothermic organisms can not regulate their body temperature.
- Respiration takes place through gill, skin and lungs. e.g. frog, toad, *Necturus*, *Icthyophis*, Salamander.
- In water frog respires through the moist skin, while on land it respires through lungs.

(ii) Reptilia

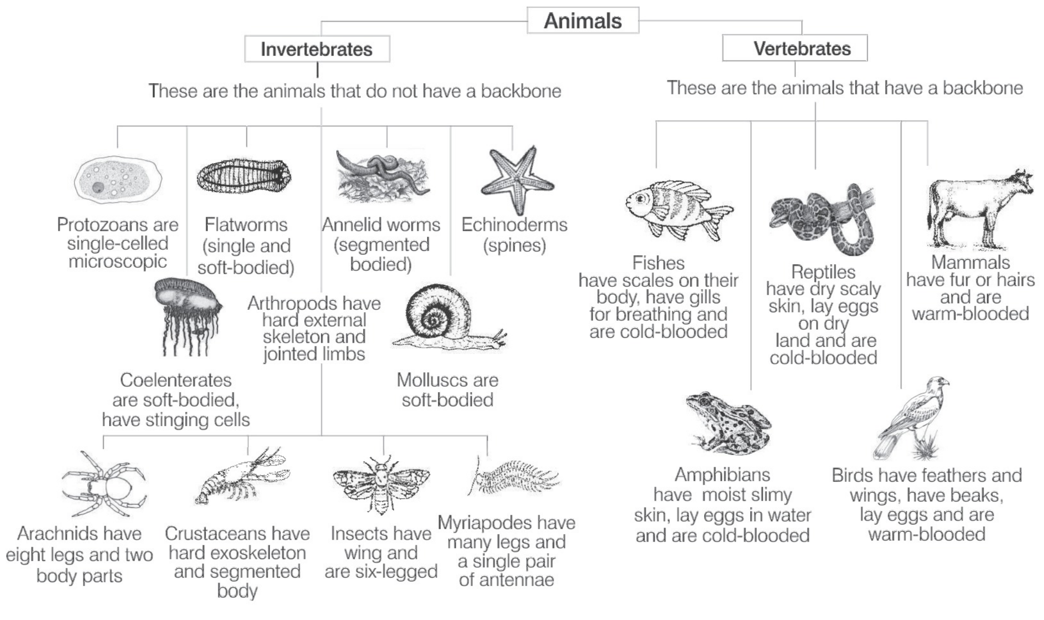
- These are crawling animals, which are cold-blooded and contain two pair of limbs.
- Their skeleton is completly flexible.
- Respiration takes place through lungs.
- Their eggs are covered with shell made up of calcium carbonate.
  e.g. lizard, snake, tortoise, crocodile, turtle, *Sphendon*, etc.
- Cobra is the only snake, which makes nests.
- Sea snake is also called *Hydrophis*. It is world’s most poisonous snake.
- *Heloderma* is the only poisonous lizard.

(iii) Aves

- The animals of this group are warm-blooded (endothermic), tetrapod vertebrates with flight adaptation and show the property of Homeothermy.
- Homeothermy is the thermoregulation that maintains internal body temperature regardless of external influence.
- Their forefeet are modified into wings for flying.
- They respire through lungs.
- Birds have no teeth, beak helps in feeding, e.g. crow, peacock, parrot, etc.
- Flightless birds are kiwi and emu. Largest alive bird is Ostrich.
- Smallest bird is humming bird.
- Largest zoo in India is Alipur (Kolkata) and the largest zoo of the world is 'Cruiser National Park' in South Africa.

(iv) Mammalia

- They are heterotrophs, mostly omnivorous organisms.
- These are warm-blooded or endothermic animals.
- Tooth comes twice in these animals (diphyodont).
- There is no nucleus in its red blood cells (except in camel and llama). Skin of mammals have hairs. External ear is present, e.g. human, whale, platypus, kangaroo, etc.
- Mammals are divided into three subclasses:
  - **Prototheria** It lays eggs, e.g. *Echidna.*
  - **Metatheria** It bears the immature child, e.g. kangaroo.
  - **Eutheria** It bears the well-developed child, e.g. humans, cow, etc.
- Mammals give birth to young ones but *Echidna* and duck-billed platypus are the only egg laying mammals.
- **Bats** have poor vision, but can fly in dark due to the production of ultrasonic waves by them.
- **Cheetah** is the fastest mammal.
- **Blue whale** is the largest mammal.
- **Spiny anteater** is the most primitive mammal.
- **Gorilla** is largest ape.
- *Panthera tigris* is the largest animal of cat family.
- **Dolphin** is the national aquatic animal of India.
- *Homo sapiens* is the commonest mammal.



Classification of animals

GENETICS AND MOLECULAR BIOLOGY AND EVOLUTION OF LIFE

GENETICS

Gregor Johann Mendel is known as the **Father of Genetics**. The term 'Genetics' was first used by **William Bateson** in 1905.

- Genetics is the process of transfer of character (traits) from one generation to next generation through genes.
- A **gene** is known as the molecular unit of heredity of a living organism.
- **Dr. Hargovind Khorana** (1968) discovered the genetic code and its function in protein systhesis.

Mendel's Experiments

GJ Mendel for the first time conducted experiments to understand the pattern of inheritance of variation in living organisms.

- He conducted hybridisation (crossing) experiments on garden pea (*Pisum sativum*) for seven years and proposed the laws of inheritance in living organisms.
- Mendel conducted artificial pollination/cross-pollination experiments using several true-breeding lines of pea.
- A **true-breeding** line refers to one that have undergone continuous self-pollination and expressed stable traits for several generations.
- Mendel selected 14 true-breeding pea plant varieties, which were similar except for one character with contrasting traits.

These characters are listed in table given below:

A List of Contrasting Traits Studied by Mendel in Pea Plant

Characters	*Contrasting traits*	
	Dominant	*Recessive*
Stem height	Tall	Dwarf
Flower colour	Violet	White
Flower position	Axial	Terminal
Pod shape	Inflated	Constricted
Pod colour	Green	Yellow
Seed shape	Round	Wrinkled
Seed colour	Yellow	Green

In 1900, Mendel's work was 'rediscovered' by three European scientists, **Hugo de Vries**, **Carl Correns** and **Erich von Tschermak**.

Mendel's Laws

Mendel's laws of inheritance are based on his observations on monohybrid and dihybrid crosses. He summarised his findings into following three laws:

Law of Dominance

(a) Law of dominance states that, when two alternative forms of a trait or character (genes) are present in an organism, only one factor expresses itself in F_1 progeny and is called **dominant**, while the other whose expression remain masked is called **recessive**.

(b) This shows that crossing between pure organisms for two contrasting characters produce only one character in F_1-generation and both in F_2-generation.

Law of Segregation

(a) This law states that, though, F_1 hybrid, alleles for dominant and recessive characters remain together, but these alleles do not show any blending and both the characters are recovered as such in the F_2-generation.

(b) These two co-existing alleles of an individual for contrasting trait segregate (separate) during gamete formation, so that each gamete receive only one of the two alleles.

(c) Alleles again unite during fertilisation of gametes.

Law of Independent Assortment

(a) It is also known as inheritance law and it states that alleles for one character assort independently of alleles for another character during gamete formation.

(b) When two pairs of traits are combined in a hybrid, segregation of one pair of character is independent of the other pair of characters during gamete formation.

➠ **Note** Back cross is performed between F_1 hybrid and one of its parent whether dominant or recessive, while test cross is a cross between F_1 hybrid and recessive parent only.

Exceptions to Principle of Dominance and Principle of Paired Factors

Incomplete dominance is the phenomenon, in which phenotype of the F_1 hybrid offspring does not resemble any of the parent, but it is an intermediate between the expression of two alleles in their homozygous state.

Red and white flowering true breeding lines of *Mirabilis jalapa* are when crossed, they shows F_1 hybrid having pink flowers, i.e. incomplete dominance (exception to dominance law).

Codominance is the phenomenon, in which two alleles of a hybrid express themselves independently. In this, offsprings show resemblance to both the parents. ABO blood group system in humans is the well-known example of codominance.